Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI/ML engineer with 6+ years of production experience, specializing in Generative AI, agentic systems, and RAG pipelines. Built two enterprise-grade AI projects end-to-end: LeaseI (8-agent LangGraph agentic workflow with Document AI/OCR pipeline, hybrid-search RAG, structured output extraction, and HITL escalation) and AgentProbe (open-source LLM evaluation toolkit with multi-provider routing, vector search, and CI regression harness). Currently integrating Anthropic Claude into financial applications at Intuit with fine-tuned prompt engineering, structured outputs, and LLM evaluation. Fine-tuned custom NLP classification models using HuggingFace + PyTorch (GMU). Strong proficiency in Python, FastAPI, Streamlit, Docker, and AWS/Azure/GCP. Case study: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Generative AI & LLMs | Agentic Systems (LangGraph) | RAG Pipelines | Document AI & OCR | Fine-tuning LLMs | Vector Databases | NLP & Deep Learning | FastAPI & Streamlit | Python (6+ yrs) | AWS / Azure / GCP | Docker & Kubernetes | Structured Output

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer - Gen AI · Virginia, USA

- Integrate Anthropic Claude into Python/FastAPI services for financial data summaries and natural-language queries; design **structured output schemas** using Pydantic so LLM responses are consistent and traceable downstream.
- Ground Claude responses in customer financial data via **retrieval pipelines** (RAG) so every answer cites a source record; add LLM-as-judge evaluation and golden-dataset CI regression tests to catch hallucinations before release.
- Build ML pipelines (scikit-learn, XGBoost) covering data ingestion, feature engineering, model training, evaluation, and **prediction API deployment** on AWS (Lambda, API Gateway).
- Add **guardrails** around LLM calls: input validation, sensitive-data filtering, fallbacks on model failure; containerize services with Docker and ship through GitHub Actions CI/CD to AWS with observability.
- Develop React/TypeScript front ends and internal tools for finance and operations teams, enabling natural-language querying of financial data.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Enterprise SaaS, 100K+ monthly active users

- Delivered 20+ features end-to-end (Python, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints in a multi-tenant enterprise platform; owned reliability of deployed services and data pipelines.
- Built Python/Bash automation pipelines for operational workflows, eliminating 80% of manual workload; optimized 200+ SQL queries (30% performance gain).
- AWS S3 migration with end-to-end Libsodium encryption for 100% PHI compliance; Docker microservices redesign (50% better scalability); Terraform for infrastructure-as-code.
- Maintained HUD income-limit data pipeline for 7 years -- ingesting, transforming, and validating ~180K domain records annually with zero compliance violations.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built application modules and data pipelines (Python, Java, PostgreSQL); delivered large-scale data reconciliation workflows (1.6M records); led AWS S3 cloud migration.

PROJECTS

Leasel -- Enterprise Agentic AI Platform with Document AI & RAG Live on GitHub

Built an enterprise-level **agentic solution** using LangGraph: 8-agent StateGraph (triage, retrieval, reasoning, validation) with HITL escalation. **Document AI/OCR pipeline**: pdfplumber + pytesseract ingestion, section-aware chunking, structured output extraction from leases and compliance documents. **Hybrid-search RAG** (hand-rolled TF-IDF + semantic reranking, no off-the-shelf LangChain) with answers cited to exact source clauses. Tested with LLM-as-judge CI harness: **93.6% precision, 87.9% F1** across 1,240 test cases. Streamlit Engineer Console + FastAPI backend.

Python · LangGraph · FastAPI · Streamlit · TF-IDF RAG · OCR (pdfplumber, pytesseract) · SQLite · Docker · CI/CD -- github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- Open-Source LLM Evaluation Toolkit Live on GitHub

Multi-provider **Gen AI platform**: 15-endpoint FastAPI backend routing across Anthropic, OpenAI, Azure OpenAI, Groq, and Ollama (both open-source and proprietary LLMs). **Vector database** layer (ChromaDB) for semantic retrieval; LLM-as-judge evaluation scoring factual accuracy, hallucination rate, retrieval quality, completeness, and compliance. Three LangGraph agents (Orchestrator, RAG Tracer, Auto Fix). CI regression pipeline on GitHub Actions with golden-dataset splits.

Python · FastAPI · LangGraph · LangChain · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama · Docker -- github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR -- NLP & LLM API Integration

Built NLP pipeline translating English/Spanish natural-language questions into FHIR API calls using LLMs, BERT embeddings, and spaCy NER -- cutting manual clinical lookup time by ~60%. Flask backend, React/TypeScript frontend, Docker, HIPAA-style RBAC.

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs -- github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University May 2026

GPA 3.87 · Fairfax, Virginia · Coursework: fine-tuned custom text classification model using HuggingFace + PyTorch on domain data, deployed via Quart API, presented to faculty.

B.Tech Computer Science & Engineering -- JNTU Anantapur 2017

TECHNICAL SKILLS

Generative AI & LLMs: Anthropic Claude, OpenAI API, Azure OpenAI, Groq, Ollama (open-source LLMs), Prompt Engineering, Fine-tuning, Structured Output, LLM-as-Judge Evaluation, Guardrails, MCP

Agentic & RAG: LangGraph, LangChain, Multi-agent Orchestration, HITL, RAG Pipelines, Hybrid Search (TF-IDF + Semantic), Embeddings, ChromaDB, Pinecone, Weaviate, Elasticsearch

Document AI & NLP: OCR (pdfplumber, pytesseract, pdf2image), Structured Output Extraction, BERT, spaCy, HuggingFace, NLP pipelines, Text Classification

ML & Deep Learning: Python, PyTorch, scikit-learn, XGBoost, PySpark, Pandas, NumPy -- supervised, unsupervised, deep learning, NLP, reinforcement learning concepts

UI & APIs: Streamlit, FastAPI, Flask, Django, REST APIs, GraphQL, Node.js, TypeScript, React, Docker, Kubernetes

Cloud & DevOps: AWS (S3, EC2, Lambda, RDS, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, AKS, Databricks), GCP (Vertex AI), Terraform, GitHub Actions CI/CD

Data & Databases: PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, SQL, Data Modeling, Query Optimization